

Supporting Information for “Fed-GEE: Federated Generalized Estimating Equations for Multisite Health Data, with an Application to Veterans’ Health”

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Sections are the Web Appendices cited in the main text. Web Tables are numbered S1, S2, and so on.

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A Large-sample theory

Let

$$\mathcal{S}_K(\beta) = \sum_{i=1}^K \mathcal{S}_{iK}(\beta), \quad \dot{\mathcal{S}}_K(\beta) = \frac{\partial \mathcal{S}_K(\beta)}{\partial \beta^\top}.$$

Theorem A.1 (Consistency and asymptotic normality). *Suppose (i) sites are independent and $E\{\mathcal{S}_{iK}(\beta_0)\} = 0$; (ii) for a deterministic continuous function s ,*

$$\sup_{\beta \in \Theta} \|K^{-1} \mathcal{S}_K(\beta) - s(\beta)\| \xrightarrow{p} 0,$$

and β_0 is a separated zero of s ; (iii), on a neighborhood of β_0 ,

$$\sup_{\beta} \|-K^{-1} \dot{\mathcal{S}}_K(\beta) - B(\beta)\| \xrightarrow{p} 0,$$

where B is continuous and $B_0 = B(\beta_0)$ is nonsingular; and (iv)

$$K^{-1/2} \mathcal{S}_K(\beta_0) \xrightarrow{d} N(0, M_0).$$

If $\hat{\beta}$ is the selected root of $\mathcal{S}_K$, then $\hat{\beta} \xrightarrow{p} \beta_0$ and

$$\sqrt{K}(\hat{\beta} - \beta_0) \xrightarrow{d} N\{0, B_0^{-1} M_0 B_0^{-\top}\}.$$

Proof. Uniform convergence and separated identification imply consistency. The integral Taylor identity gives

$$0 = \mathcal{S}_K(\beta_0) + \left\{ \int_0^1 \dot{\mathcal{S}}_K(\beta_0 + t(\hat{\beta} - \beta_0)) dt \right\} (\hat{\beta} - \beta_0).$$

The negative normalized integral Jacobian converges in probability to B_0 . Premultiplication by its inverse followed by Slutsky's theorem gives the stated limit. $\square$

B Exactness of the half-power

All expectations in this section are conditional on the design.

Theorem B.1 (Exact score-space correction). *Suppose*

$$\mathcal{S}_i(\beta) = U_i - B_i(\beta - \beta_0),$$

where B_i is fixed and positive semidefinite, the U_i are independent, $E(U_i) = 0$, and $\text{Var}(U_i) = \tau B_i$ for a common $\tau > 0$. Let $B = \sum_i B_i \succ 0$ and assume $B - B_i \succ 0$ for every i . Define

$$G_i = B^{-1/2} B_i B^{-1/2}, \quad z_i = B^{-1/2} \mathcal{S}_i(\hat{\beta}), \quad A_i = (\mathbf{I} - G_i)^{-1/2}.$$

Then

$$E(z_i z_i^\top) = \tau G_i (\mathbf{I} - G_i)$$

and

$$\hat{V}_{\text{KC}} = B^{-1/2} \left\{ \sum_i A_i z_i z_i^\top A_i \right\} B^{-1/2}$$

satisfies

$$E(\hat{V}_{\text{KC}}) = \tau B^{-1} = \text{Var}(\hat{\beta}).$$

Proof. The estimating equation gives

$$\hat{\beta} - \beta_0 = B^{-1} \sum_j U_j.$$

Writing $H_i = B_i B^{-1}$,

$$\mathcal{S}_i(\hat{\beta}) = (\mathbf{I} - H_i) U_i - H_i \sum_{j \neq i} U_j.$$

Independence and the covariance condition therefore imply

$$\begin{aligned} E\{\mathcal{S}_i(\hat{\beta}) \mathcal{S}_i(\hat{\beta})^\top\} &= (\mathbf{I} - H_i)(\tau B_i)(\mathbf{I} - H_i)^\top \\ &\quad + H_i\{\tau(B - B_i)\}H_i^\top. \end{aligned}$$

Conjugation by $B^{-1/2}$ yields

$$E(z_i z_i^\top) = \tau G_i (\mathbf{I} - G_i).$$

Because A_i is a matrix function of G_i , the two commute, and

$$A_i E(z_i z_i^\top) A_i = \tau G_i.$$

Finally, $\sum_i G_i = \mathbf{I}$, so

$$E(\hat{V}_{\text{KC}}) = B^{-1/2} \left(\tau \sum_i G_i \right) B^{-1/2} = \tau B^{-1}.$$

The same value follows directly for the variance of $B^{-1} \sum_i U_i$. □

B.1 Uniqueness of the half-power

For $A_i(\alpha) = (\mathbf{I} - G_i)^{-\alpha}$, the expected adjusted information-standardized contribution is

$$\tau G_i (\mathbf{I} - G_i)^{1-2\alpha}.$$

Requiring this to equal τG_i for every admissible G_i forces $1 - 2\alpha = 0$. Thus $\alpha = 1/2$ is unique within this family. The full-power choice $\alpha = 1$ is larger in positive-semidefinite order under the affine working model and is consequently expected to be conservative.

B.2 Raw-score representation

The equivalent adjusted raw score is

$$\tilde{\mathcal{S}}_i = B^{1/2} (\mathbf{I} - G_i)^{-1/2} B^{-1/2} \mathcal{S}_i(\hat{\beta}),$$

and

$$\hat{V}_{\text{KC}} = B^{-1} \left(\sum_i \tilde{\mathcal{S}}_i \tilde{\mathcal{S}}_i^\top \right) B^{-1}.$$

This representation shows why the correction cannot generally be implemented by multiplying a raw score directly by $(\mathbf{I} - G_i)^{-1/2}$.

C Invariance to the matrix square root

All three corrections in the main text use the symmetric square root $\hat{\mathcal{B}}^{-1/2}$. Any other factor gives the same variance.

Proposition C.1 (Invariance to the choice of matrix factor). *Let $B \succ 0$ and let R be any $p \times p$ matrix with $RR^\top = B^{-1}$. Define*

$$G_i^R = R^\top B_i R, \quad z_i^R = R^\top \mathcal{S}_i(\hat{\beta}), \quad A_i^R(\alpha) = (\mathbf{I} - G_i^R)^{-\alpha}.$$

Then, for every $\alpha > 0$,

$$R \left\{ \sum_{i=1}^K A_i^R(\alpha) z_i^R z_i^{R\top} A_i^R(\alpha)^\top \right\} R^\top = B^{-1/2} \left\{ \sum_{i=1}^K A_i(\alpha) z_i z_i^\top A_i(\alpha) \right\} B^{-1/2}.$$

Proof. Let $Q = B^{1/2} R$. Then $QQ^\top = B^{1/2} B^{-1} B^{1/2} = \mathbf{I}$, so Q is orthogonal and $R = B^{-1/2} Q$. Hence $G_i^R = Q^\top G_i Q$ and $z_i^R = Q^\top z_i$. Because Q is orthogonal, $(\mathbf{I} - Q^\top G_i Q)^{-\alpha} =$

$Q^\top(\mathbf{I} - G_i)^{-\alpha}Q$. Substituting, each term equals $B^{-1/2}QQ^\top A_iQQ^\top z_i z_i^\top QQ^\top A_iQQ^\top B^{-1/2} = B^{-1/2}A_i z_i z_i^\top A_i B^{-1/2}$. $\square$

A Cholesky factor of B^{-1} may therefore be used in computation. The same argument shows that the degrees of freedom of Web Appendix F do not depend on the factor, because they depend on the design only through the corrected raw-score operators T_i , which are factor invariant.

D Large- K equivalence of the corrections

Proposition D.1 (Large- K equivalence of the corrections). *Let $B_K = \hat{\mathcal{B}}$, $q_K = \max_i \|G_i\|$, and $z_i = B_K^{-1/2}\hat{\mathcal{S}}_i$. Suppose $q_K = O_p(K^{-1})$, $\sum_i \|z_i\|^2 = O_p(1)$, and $\|B_K^{-1/2}\| = O_p(K^{-1/2})$. Then $\|\hat{V}_{\text{KC}} - \hat{V}_S\| = O_p(K^{-2})$, while $\|\hat{V}_S\| = O_p(K^{-1})$.*

The condition $q_K = O_p(K^{-1})$ says that no site holds more than a vanishing share of the information.

Proof. Let

$$A_i(1/2) = (\mathbf{I}_p - G_i)^{-1/2}, \quad q_K = \max_{1 \leq i \leq K} \|G_i\|.$$

The scalar expansion

$$(1 - x)^{-1/2} = \sum_{m=0}^{\infty} c_m x^m, \quad c_m = \frac{\binom{2m}{m}}{4^m},$$

induces a convergent matrix power series whenever $\|G_i\| < 1$. Because $q_K = O_p(K^{-1})$, with probability tending to one $q_K \leq 1/2$. On this event,

$$d_K := \max_i \|A_i(1/2) - \mathbf{I}_p\| \leq \sum_{m=1}^{\infty} c_m q_K^m = (1 - q_K)^{-1/2} - 1 \leq C q_K$$

for a finite constant C . Hence

$$d_K = O_p(K^{-1}), \quad \max_i \|A_i(1/2)\| = O_p(1).$$

Using the identity

$$\begin{aligned} & A_i(1/2) z_i z_i^\top A_i(1/2) - z_i z_i^\top \\ &= \{A_i(1/2) - \mathbf{I}_p\} z_i z_i^\top A_i(1/2) + z_i z_i^\top \{A_i(1/2) - \mathbf{I}_p\}, \end{aligned}$$

submultiplicativity and $\|z_i z_i^\top\| = \|z_i\|^2$ give

$$\begin{aligned} & \left\| \sum_{i=1}^K \left(A_i(1/2) z_i z_i^\top A_i(1/2) - z_i z_i^\top \right) \right\| \\ & \leq d_K \left\{ \max_i \|A_i(1/2)\| + 1 \right\} \sum_{i=1}^K \|z_i\|^2 = O_p(K^{-1}). \end{aligned}$$

Therefore, using $\|B_K^{-1/2}\| = O_p(K^{-1/2})$,

$$\|\hat{V}_{\text{KC}} - \hat{V}_{\text{S}}\| \leq \|B_K^{-1/2}\|^2 O_p(K^{-1}) = O_p(K^{-2}).$$

Moreover,

$$\|\hat{V}_{\text{S}}\| \leq \|B_K^{-1/2}\|^2 \sum_{i=1}^K \|z_i\|^2 = O_p(K^{-1}).$$

Thus the half-power correction is second order relative to the K^{-1} scale of the conventional sandwich covariance and does not alter its first-order large- K limit. $\square$

Remark D.2. The same argument applies to MD, with the coefficients c_m replaced by 1, and to FG, whose diagonal factors satisfy $\|F_i - \mathbf{I}\| = O_p(K^{-1})$ when the condition number of $K^{-1}B_K$ is bounded. All corrections therefore share the first-order limit of the uncorrected sandwich.

E Rank of the fitted meat

For the unadjusted fitted score matrix $S = [\mathcal{S}_1(\hat{\beta}), \dots, \mathcal{S}_K(\hat{\beta})]$, the estimating equation gives $S\mathbf{1}_K = 0$. Consequently,

$$\text{rank}(SS^\top) \leq \min(p, K - 1).$$

Site-specific invertible adjustments need not preserve the zero-sum relation. The universal adjusted bound is instead

$$\text{rank} \left(\sum_i A_i \mathcal{S}_i \mathcal{S}_i^\top A_i^\top \right) \leq \min(p, K).$$

Thus an adjusted meat is necessarily singular when $K < p$, but it can be full rank when $K = p$. This distinction does not rescue routine Wald inference at $K \leq p$, where the uncorrected score covariance and residual degrees of freedom are inadequate.

F Fay–Graubard and Bell–McCaffrey in score space

F.1 Fay–Graubard in score space

Fay and Graubard [1] rescale each cluster’s contribution to the meat by a diagonal factor built from the cluster’s share of the information. At the site level this factor is

$$F_i = \text{diag} \left[\left\{ 1 - \min \left(b, (B_i B^{-1})_{kk} \right) \right\}^{-1/2} \right]_k^p,$$

with $b = 0.75$ by default. The corrected variance is $B^{-1} \left(\sum_i F_i \hat{\mathcal{S}}_i \hat{\mathcal{S}}_i^\top F_i \right) B^{-1}$. It uses only the site scores and information matrices. Our implementation agreed with the **saws** package (method **d5**) to machine precision when **saws** was given the same site scores and information matrices.

F.2 Bell–McCaffrey degrees of freedom in score space

Fix a coefficient j and a correction with raw-score operators T_i (main text, equation (9)). Let $a_i = T_i^\top B^{-1} e_j$. The corrected variance of $\hat{\beta}_j$ is

$$v_j = \sum_{i=1}^K (a_i^\top \hat{\mathcal{S}}_i)^2.$$

Work under the affine model of Theorem 1 of the main text, with $U_k = \tau^{1/2} B_k^{1/2} \varepsilon_k$ and independent standard normal ε_k . Then $\hat{\mathcal{S}}_i = U_i - B_i B^{-1} \sum_k U_k$, so

$$a_i^\top \hat{\mathcal{S}}_i = \sum_{k=1}^K w_{ik}^\top U_k, \quad w_{ik} = \mathbf{1}\{i = k\} a_i - B^{-1} B_i a_i.$$

Hence $v_j = \tau \varepsilon^\top Q \varepsilon$ for a positive semidefinite matrix $Q = \sum_i r_i r_i^\top$, where r_i stacks the blocks $B_k^{1/2} w_{ik}$. The nonzero eigenvalues of Q equal those of the $K \times K$ matrix C with entries $C_{il} = r_i^\top r_l = \sum_k w_{ik}^\top B_k w_{lk}$. Expanding and using $\sum_k B_k = B$,

$$C_{il} = \mathbf{1}\{i = l\} a_i^\top B_i a_i - (B_i a_i)^\top B^{-1} (B_l a_l).$$

For a Gaussian quadratic form, $E(v_j) = \tau \text{tr}(C)$ and $\text{Var}(v_j) = 2\tau^2 \text{tr}(C^2)$. Matching these two moments to a scaled χ_ν^2 distribution [2, 3] gives

$$\nu_j^{\text{BM}} = \frac{\{\text{tr}(C)\}^2}{\text{tr}(C^2)}.$$

Equal sites. If $B_i = B/K$ and $T_i = T$ for all i , then $a_i = a$ and $C = c(\mathbf{I}_K - K^{-1}\mathbf{1}_K\mathbf{1}_K^\top)$ with $c = a^\top Ba/K$. Then $\text{tr}(C) = c(K-1)$ and $\text{tr}(C^2) = c^2(K-1)$, so $\nu_j^{\text{BM}} = K-1$.

Relation to CR2. With working independence and an identity link, the site information matrices are $X_i^\top X_i$, the KC operator equals the CR2 adjustment, and ν_j^{BM} equals the Satterthwaite degrees of freedom of Pustejovsky and Tipton [4]. We confirmed agreement to four decimal places against the `clubSandwich` package in simulated data. In the VHA application, with a logistic link, the degrees of freedom agreed to three decimal places and the standard errors to within 0.5%.

Monte Carlo check. Under the working model with 20,000 simulated draws, the empirical mean and variance of v_j matched $\tau \text{tr}(C)$ and $2\tau^2 \text{tr}(C^2)$.

G One-step Fed-GEE

When communication rounds are costly, a common initial estimator $\hat{\beta}^{(0)}$ can be updated once using

$$\hat{\beta}^{(1)} = \hat{\beta}^{(0)} + \mathcal{B}_K\{\hat{\beta}^{(0)}\}^{-1} \mathcal{S}_K\{\hat{\beta}^{(0)}\}. \quad (\text{G.1})$$

Each site sends one score and one information matrix evaluated at the common initializer.

Assumption G.1 (Initial rate). *The common initializer satisfies*

$$\hat{\beta}^{(0)} - \beta_0 = o_p(K^{-1/4}).$$

Assumption G.2 (Local smoothness). *In a neighborhood of β_0 ,*

$$\|B_i(\beta) - B_i(\beta')\| \leq \ell_i \|\beta - \beta'\|, \quad K^{-1} \sum_i \ell_i = O_p(1),$$

and the normalized difference between $B_K(\beta)$ and the negative score derivative is $O_p(K^{-1/2})$ uniformly in that neighborhood.

Theorem G.1 (One-step Fed-GEE). *Under the centralized conditions and the two preceding assumptions,*

$$\sqrt{K}(\hat{\beta}^{(1)} - \beta_0) \xrightarrow{d} N\{0, B_0^{-1} M_0 B_0^{-\top}\}.$$

Thus the one-step and fully converged estimators have the same first-order limit.

Proof. Expand $\mathcal{S}_K\{\hat{\beta}^{(0)}\}$ around β_0 using an integral Jacobian. After substituting into Equation (G.1), the linear term cancels the initial error. The bread Lipschitz condition

bounds the remaining curvature term by

$$O_p(1)\sqrt{K}\left\|\hat{\beta}^{(0)} - \beta_0\right\|^2 = o_p(1).$$

The normalized bread–score-derivative discrepancy contributes $O_p(1)\left\|\hat{\beta}^{(0)} - \beta_0\right\| = o_p(1)$. Hence

$$\{K^{-1}\mathcal{B}_K(\hat{\beta}^{(0)})\}\sqrt{K}(\hat{\beta}^{(1)} - \beta_0) = K^{-1/2}\mathcal{S}_K(\beta_0) + o_p(1).$$

The bread converges to B_0 , and the site-level central limit theorem and Slutsky’s theorem give the result. $\square$

The rate condition is substantive. A one-shot update is not recommended when a common $o_p(K^{-1/4})$ initializer cannot be justified.

H Decentralized Fed-GEE

H.1 Communication Conditions

In the decentralized architecture, each site exchanges low-dimensional quantities only with its network neighbors and approximates global summaries through average consensus. The centralized regularity conditions stated in the main manuscript continue to govern the statistical model; the conditions below concern communication and numerical approximation.

Condition H.1 (Communication network). *For each K , let*

$$W_K = (w_{ih}) \in \mathbb{R}^{K \times K}$$

denote the mixing matrix, where i and h index sites. The entries of W_K are nonnegative, and W_K is symmetric and doubly stochastic:

$$W_K \mathbf{1}_K = \mathbf{1}_K, \quad \mathbf{1}_K^\top W_K = \mathbf{1}_K^\top.$$

The corresponding undirected communication graph is connected, with

$$w_{ih} = 0$$

whenever sites i and h are not neighbors and $i \neq h$. Define

$$J_K = \frac{1}{K} \mathbf{1}_K \mathbf{1}_K^\top.$$

Assume the mixing process is aperiodic, for example through positive diagonal weights, so that

$$\rho_K = \|W_K - J_K\|_2 < 1.$$

For a site-level quantity Q , define

$$\delta_{Q,K} = \|W_K^{L_{Q,K}} - J_K\|_\infty, \quad (\text{H.1})$$

where $L_{Q,K}$ is the number of consensus rounds applied to Q .

Condition H.2 (Site-level smoothness and boundedness). For all $\beta_1, \beta_2 \in \Theta$,

$$\|S_h(\beta_1) - S_h(\beta_2)\|_2 \leq \ell_{S,h,K} \|\beta_1 - \beta_2\|_2,$$

where

$$\frac{1}{K} \sum_{h=1}^K \ell_{S,h,K} = O_p(1).$$

Similarly,

$$\|B_h(\beta_1) - B_h(\beta_2)\|_2 \leq \ell_{B,h,K} \|\beta_1 - \beta_2\|_2,$$

where

$$\frac{1}{K} \sum_{h=1}^K \ell_{B,h,K} = O_p(1).$$

In addition,

$$\max_h \sup_{\beta \in \Theta} \|S_h(\beta)\|_2 = O_p(1), \quad \max_h \sup_{\beta \in \Theta} \|B_h(\beta)\|_2 = O_p(1).$$

Let $\beta_h^\star$ denote the terminal local parameter state at site h immediately before the final parameter-consensus step. Assume

$$\max_h \|\beta_h^\star\|_2 = O_p(1).$$

Condition H.3 (Terminal algorithmic accuracy). After the final parameter-consensus step, site i holds

$$\hat{\beta}_i = \sum_{h=1}^K [W_K^{L_{\beta,K}}]_{ih} \beta_h^\star. \quad (\text{H.2})$$

At $\hat{\beta}_h$, each site recomputes its local score and bread. After score and bread consensus, site i holds

$$\hat{S}_i = \sum_{h=1}^K [W_K^{L_{S,K}}]_{ih} S_h(\hat{\beta}_h),$$

and

$$\hat{\hat{B}}_i = \sum_{h=1}^K \left[W_K^{L_{B,K}} \right]_{ih} B_h(\hat{\beta}_h).$$

Assume $\hat{\hat{B}}_i$ is nonsingular with probability tending to one. Define the achieved Fisher-scoring residual by

$$\eta_K = \max_i \left\| \hat{\hat{B}}_i^{-1} \hat{\hat{S}}_i \right\|_2. \quad (\text{H.3})$$

The algorithm is terminated only when

$$\eta_K \leq \epsilon_{\eta,K}.$$

For consistency, require $\epsilon_{\eta,K} \rightarrow 0$; for root- K asymptotic equivalence, require

$$\epsilon_{\eta,K} = o(K^{-1/2}).$$

H.2 Estimator and Consensus Approximation

Define the decentralized estimator by the network average

$$\hat{\beta}_{\text{dec}} = \frac{1}{K} \sum_{i=1}^K \hat{\beta}_i. \quad (\text{H.4})$$

Because W_K is doubly stochastic,

$$\hat{\beta}_{\text{dec}} = \frac{1}{K} \sum_{h=1}^K \beta_h^*.$$

Thus the final parameter-consensus step preserves the network average.

Lemma H.1 (Average-consensus approximation). *Let $Q_1, \dots, Q_K$ be site-specific vectors or matrices satisfying*

$$\max_h \|Q_h\| = O_p(1).$$

After $L_{Q,K}$ consensus rounds, let

$$\hat{\hat{Q}}_i = \sum_{h=1}^K \left[W_K^{L_{Q,K}} \right]_{ih} Q_h.$$

Then

$$\max_i \left\| \hat{\hat{Q}}_i - \frac{1}{K} \sum_{h=1}^K Q_h \right\| = O_p(\delta_{Q,K}).$$

Lemma H.2 (Final parameter agreement). *Under Conditions H.1 and H.2,*

$$\max_i \left\| \hat{\beta}_i - \hat{\beta}_{\text{dec}} \right\|_2 = O_p(\delta_{\beta,K}).$$

Lemma H.3 (Approximate global estimating equation). *Under Conditions H.1–H.3,*

$$\left\| \bar{S}_K(\hat{\beta}_{\text{dec}}) \right\|_2 = O_p(r_K),$$

where

$$r_K = \delta_{\beta,K} + \delta_{S,K} + \epsilon_{\eta,K}. \quad (\text{H.5})$$

H.3 Large-Sample Properties

Theorem H.4 (Consistency and asymptotic equivalence). *Suppose the centralized consistency conditions stated in the main manuscript hold, together with Conditions H.1– H.3. If*

$$r_K = o(1),$$

then

$$\hat{\beta}_{\text{dec}} \xrightarrow{p} \beta_0.$$

Moreover,

$$\left\| \hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}} \right\|_2 = O_p(r_K).$$

Theorem H.5 (Asymptotic normality). *Suppose the centralized asymptotic-normality conditions stated in the main manuscript hold, together with Conditions H.1– H.3. If*

$$r_K = o(K^{-1/2}),$$

then

$$\sqrt{K}(\hat{\beta}_{\text{dec}} - \beta_0) \xrightarrow{d} N\left(\mathbf{0}, B_0^{-1} M_0 B_0^{-\top}\right).$$

Corollary H.6 (Sufficient communication and stopping rates). *Consistency follows if*

$$\delta_{\beta,K} \rightarrow 0, \quad \delta_{S,K} \rightarrow 0, \quad \epsilon_{\eta,K} \rightarrow 0.$$

Root-K asymptotic equivalence follows if

$$\delta_{\beta,K} + \delta_{S,K} + \epsilon_{\eta,K} = o(K^{-1/2}).$$

If the final global bread is also reconstructed by consensus, require

$$\delta_{B,K} \rightarrow 0.$$

For the finite-sample score correction, the fitted leverage matrices must remain well defined, with

$$\lambda_{\max}(\hat{G}_i) < 1$$

in parameter directions in which a negative matrix power is used.

H.4 Consensus Rounds

For any site-level quantity Q_i , let $Q_i^{[0]} = Q_i$. At round ℓ ,

$$Q_i^{[\ell+1]} = \sum_{h=1}^K w_{ih} Q_h^{[\ell]}.$$

Stacking the site states gives

$$Q^{[\ell+1]} = W_K Q^{[\ell]},$$

and after L rounds,

$$Q^{[L]} = W_K^L Q^{[0]}.$$

For a symmetric mixing matrix,

$$\|W_K^L - J_K\|_2 = \rho_K^L, \quad \rho_K = \|W_K - J_K\|_2 < 1.$$

Because, for any $K \times K$ matrix C ,

$$\|C\|_\infty \leq \sqrt{K} \|C\|_2,$$

the row-sum consensus error satisfies

$$\|W_K^L - J_K\|_\infty \leq \sqrt{K} \rho_K^L. \tag{H.6}$$

Thus increasing the number of communication rounds reduces the worst-case consensus bound geometrically. The realized error at an individual site need not decrease monotonically after every single additional round because network eigen-components can temporarily cancel or reinforce one another.

H.5 Finite-Sample Score Correction in the Decentralized Architecture

After final parameter consensus, site i evaluates

$$S_i(\hat{\beta}_i), \quad B_i(\hat{\beta}_i).$$

After L rounds of bread consensus, let

$$\hat{B}_i^{(L)} = \sum_{h=1}^K [W_K^L]_{ih} B_h(\hat{\beta}_h),$$

and reconstruct the network-wide bread sum by

$$\hat{B}_{K,i}^{(L)} = K \hat{B}_i^{(L)}. \quad (\text{H.7})$$

Site i then forms

$$\hat{G}_i^{(L)} = \{\hat{B}_{K,i}^{(L)}\}^{-1/2} B_i(\hat{\beta}_i) \{\hat{B}_{K,i}^{(L)}\}^{-1/2}, \quad (\text{H.8})$$

and, for $a \in \{0, 1/2, 1\}$,

$$\hat{A}_i^{(L)}(a) = (\mathbf{I}_p - \hat{G}_i^{(L)})^{-a}. \quad (\text{H.9})$$

The corrected local score is

$$S_i^{\text{corr},(L)}(a) = \{\hat{B}_{K,i}^{(L)}\}^{1/2} \hat{A}_i^{(L)}(a) \{\hat{B}_{K,i}^{(L)}\}^{-1/2} S_i(\hat{\beta}_i), \quad (\text{H.10})$$

with local corrected meat contribution

$$M_i^{\text{corr},(L)}(a) = S_i^{\text{corr},(L)}(a) S_i^{\text{corr},(L)}(a)^\top.$$

These local meat contributions can be aggregated through an additional consensus step.

Let

$$\hat{B}_K^{\text{dec}} = \sum_{h=1}^K B_h(\hat{\beta}_h).$$

By Lemma H.1,

$$\left\| \hat{B}_i^{(L)} - \frac{1}{K} \hat{B}_K^{\text{dec}} \right\| = O_p(\delta_{B,K}^{(L)}),$$

so

$$\left\| \hat{B}_{K,i}^{(L)} - \hat{B}_K^{\text{dec}} \right\| = O_p\{K \delta_{B,K}^{(L)}\}.$$

Since the global bread is of order K , this corresponds to a relative error of order $O_p(\delta_{B,K}^{(L)})$.

Under continuity of the inverse square-root and matrix-power maps, together with eigenvalue gaps away from zero, vanishing parameter- and bread-consensus errors imply that $\hat{G}_i^{(L)}$, $\hat{A}_i^{(L)}(a)$, and the corrected local meat contributions converge to their centralized counterparts.

H.5.1 Reference distributions from additive summaries

The Bell–McCaffrey degrees of freedom also admit a consensus formulation. For a fixed coefficient j , write $B = \sum_i B_i$, $a_i = T_i^\top B^{-1} e_j$, $d_i = a_i^\top B_i a_i$, and $u_i = B_i a_i$. Define the additive summaries

$$D_1 = \sum_i d_i, \quad D_2 = \sum_i d_i^2, \quad H = \sum_i u_i u_i^\top, \quad R = \sum_i d_i u_i u_i^\top.$$

The kernel in the main text is $C = D - U^\top B^{-1} U$, where $D = \text{diag}(d_1, \dots, d_K)$ and U has columns u_i . Cyclic invariance of the trace gives

$$\text{tr}(C) = D_1 - \text{tr}(B^{-1} H), \tag{H.11}$$

$$\text{tr}(C^2) = D_2 - 2 \text{tr}(B^{-1} R) + \text{tr}(B^{-1} H B^{-1} H). \tag{H.12}$$

Thus, no exchange of the full $K \times K$ kernel is required. After information consensus, each site constructs its local d_i, u_i using its approximation to B , and applies consensus to $d_i, d_i^2, u_i u_i^\top, d_i u_i u_i^\top$. Multiplication by K converts the consensus averages into approximate totals.

For fixed K , continuity of these expressions implies convergence to the coordinated degrees of freedom as parameter and summary consensus errors vanish, provided the information matrix remains nonsingular, the correction operators remain well defined, and $\text{tr}(C^2) > 0$ at the limit. The resulting critical values converge whenever the limiting degrees of freedom are positive. This is a computational equivalence result and does not establish exact finite-sample coverage. The decentralized simulations in Web Appendix I do not evaluate this extension.

H.6 Proofs for the Decentralized Results

H.6.1 Proof of Lemma H.1

Proof. For every site i ,

$$\hat{\bar{Q}}_i - \frac{1}{K} \sum_{h=1}^K Q_h = \sum_{h=1}^K \left\{ [W_K^{L_{Q,K}}]_{ih} - \frac{1}{K} \right\} Q_h.$$

By the triangle inequality,

$$\left\| \hat{\bar{Q}}_i - \frac{1}{K} \sum_{h=1}^K Q_h \right\| \leq \|W_K^{L_{Q,K}} - J_K\|_{\infty} \max_h \|Q_h\|.$$

Taking the maximum over i gives

$$\max_i \left\| \hat{\bar{Q}}_i - \frac{1}{K} \sum_{h=1}^K Q_h \right\| = O_p(\delta_{Q,K}).$$

□

H.6.2 Proof of Lemma H.2

Proof. Because W_K is doubly stochastic,

$$\hat{\beta}_{\text{dec}} = \frac{1}{K} \sum_{h=1}^K \beta_h^*.$$

Applying Lemma H.1 with $Q_h = \beta_h^*$ gives

$$\max_i \left\| \hat{\beta}_i - \hat{\beta}_{\text{dec}} \right\|_2 = O_p(\delta_{\beta,K}).$$

□

H.6.3 Proof of Lemma H.3

Proof. Fix a site i and decompose

$$\bar{S}_K(\hat{\beta}_{\text{dec}}) = R_{S,1} + R_{S,2,i} + R_{S,3,i},$$

where

$$R_{S,1} = \frac{1}{K} \sum_{h=1}^K \{S_h(\hat{\beta}_{\text{dec}}) - S_h(\hat{\beta}_h)\},$$

$$R_{S,2,i} = \frac{1}{K} \sum_{h=1}^K S_h(\hat{\beta}_h) - \hat{S}_i, \quad R_{S,3,i} = \hat{S}_i.$$

By site-level score Lipschitz continuity and Lemma H.2,

$$\|R_{S,1}\|_2 \leq \left(\frac{1}{K} \sum_{h=1}^K \ell_{S,h,K} \right) \max_h \|\hat{\beta}_h - \hat{\beta}_{\text{dec}}\|_2 = O_p(\delta_{\beta,K}).$$

Lemma H.1 applied to $Q_h = S_h(\hat{\beta}_h)$ gives

$$\max_i \|R_{S,2,i}\|_2 = O_p(\delta_{S,K}).$$

For the final term,

$$\hat{S}_i = \hat{B}_i(\hat{B}_i^{-1} \hat{S}_i).$$

The bounded-bread condition and convexity of the consensus weights give

$$\max_i \|\hat{B}_i\|_2 = O_p(1),$$

while the stopping rule gives

$$\max_i \|\hat{B}_i^{-1} \hat{S}_i\|_2 \leq \epsilon_{\eta,K}.$$

Hence

$$\max_i \|R_{S,3,i}\|_2 = O_p(\epsilon_{\eta,K}).$$

Combining terms yields

$$\|\bar{S}_K(\hat{\beta}_{\text{dec}})\|_2 = O_p(\delta_{\beta,K} + \delta_{S,K} + \epsilon_{\eta,K}) = O_p(r_K).$$

□

H.6.4 Proof of Theorem H.4

Proof. Lemma H.3 gives

$$\|\bar{S}_K(\hat{\beta}_{\text{dec}})\|_2 = O_p(r_K) = o_p(1).$$

Let $s(\beta)$ denote the deterministic uniform limit of $\bar{S}_K(\beta)$. Then

$$\|s(\hat{\beta}_{\text{dec}})\|_2 \leq \sup_{\beta \in \Theta} \|s(\beta) - \bar{S}_K(\beta)\|_2 + \|\bar{S}_K(\hat{\beta}_{\text{dec}})\|_2 = o_p(1).$$

Identification therefore implies

$$\hat{\beta}_{\text{dec}} \xrightarrow{p} \beta_0.$$

Since $\bar{S}_K(\hat{\beta}_{\text{cen}}) = \mathbf{0}$, a first-order expansion between the centralized and decentralized estimators gives

$$\bar{S}_K(\hat{\beta}_{\text{dec}}) = -\bar{B}_K^*(\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}) + o_p(\|\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}\|),$$

where $\bar{B}_K^* \xrightarrow{p} B_0$. Therefore,

$$\|\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}\|_2 = O_p(r_K).$$

□

H.6.5 Proof of Theorem H.5

Proof. From Theorem H.4,

$$\|\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}\|_2 = O_p(r_K).$$

If $r_K = o(K^{-1/2})$, then

$$\sqrt{K}(\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}) = o_p(1).$$

Therefore,

$$\sqrt{K}(\hat{\beta}_{\text{dec}} - \beta_0) = \sqrt{K}(\hat{\beta}_{\text{cen}} - \beta_0) + o_p(1).$$

The centralized asymptotic-normality result and Slutsky's theorem give

$$\sqrt{K}(\hat{\beta}_{\text{dec}} - \beta_0) \xrightarrow{d} N(\mathbf{0}, B_0^{-1} M_0 B_0^{-\top}).$$

□

I Decentralized simulation

This appendix reports a simulation of the decentralized algorithm of Web Appendix H. It was run before the correction and reference distribution of the main text were finalized. All intervals use the uncorrected site sandwich with a t_{K-p} reference. The simulation therefore shows how consensus error affects estimation and inference. It does not evaluate the KC correction or the Bell–McCaffrey degrees of freedom under consensus.

Design. Outcomes followed the latent Gaussian design of the main text, with patient-level correlation 0.30 and site-level correlation $\rho_H \in \{0, 0.1, 0.2\}$. Sites had 10 to 30 patients with 2 to 6 visits each. The number of sites was $K \in \{5, 10, 15, 25, 50, 100\}$, the average prevalence was 0.07 or 0.30, and some scenarios added a binary site-level covariate ($p = 3$). Each scenario had 200 to 400 replicates. Four networks were compared: a complete graph; a

regional network, in which sites form groups of five around a regional hub and the hubs are fully connected; a ring in which each site is linked to its two nearest neighbors on each side; and a single hub linked to every site. Mixing weights were Metropolis–Hastings weights. Each consensus stage used $L = 50$ rounds.

Results. Web Table S1 gives the spectral quantity ρ that governs consensus error, and Web Table S2 gives coverage and SE/SD by scenario. On the complete graph the decentralized results equalled the centralized results in every scenario. On the regional network they differed by at most 0.01 in coverage. The hub network matched the centralized results for $K \leq 25$, where $\rho^{50} \leq 0.13$. At $K = 50$ its coverage fell to 0.795 to 0.913, and at $K = 100$, where $\rho^{50} = 0.61$, to 0.060 to 0.580. The estimate stayed within 0.1 of the true slope of 1, but this error was up to 2.9 standard deviations, and the standard errors were too small (SE/SD 0.51 to 0.63). The ring had $\rho^{50} = 0.94$ at $K = 100$ and covered at 0.905 to 0.953, compared with 0.925 to 0.965 centrally. A fixed number of rounds is therefore adequate for some networks and not for others. The required number depends on ρ and grows as the network becomes less connected.

Table S1: Second-largest eigenvalue modulus ρ of the mixing matrix, and ρ^{50} , for each network and number of sites K .

K	Complete		Regional		Ring		Hub	
	ρ	ρ^{50}	ρ	ρ^{50}	ρ	ρ^{50}	ρ	ρ^{50}
5	0.000	0.000	0.000	0.000	0.539	0.000	0.800	0.000
10	0.000	0.000	0.950	0.078	0.873	0.001	0.900	0.005
15	0.000	0.000	0.944	0.055	0.942	0.051	0.933	0.032
25	0.000	0.000	0.941	0.049	0.979	0.347	0.960	0.130
50	0.000	0.000	0.950	0.077	0.995	0.768	0.980	0.364
100	0.000	0.000	0.966	0.173	0.999	0.936	0.990	0.605

Table S2: Coverage of nominal 95% t_{K-p} intervals and SE/SD for β_x , centralized Fed-GEE and decentralized Fed-GEE with $L = 50$ consensus rounds, by scenario. Uncorrected site sandwich throughout.

K	ρ_H	Prev.	p	Coverage					SE/SD				
				Cen.	Comp.	Reg.	Ring	Hub	Cen.	Comp.	Reg.	Ring	Hub
5	0.0	0.30	2	0.953	0.953	0.953	0.953	0.953	0.791	0.791	0.791	0.791	0.791
5	0.0	0.07	2	0.940	0.940	0.940	0.940	0.940	0.820	0.820	0.820	0.820	0.820
5	0.1	0.30	2	0.930	0.930	0.930	0.930	0.930	0.762	0.762	0.762	0.762	0.762
5	0.1	0.07	2	0.925	0.925	0.925	0.925	0.925	0.753	0.753	0.753	0.753	0.753
5	0.2	0.30	2	0.940	0.940	0.940	0.940	0.940	0.778	0.778	0.778	0.778	0.778
5	0.2	0.07	2	0.925	0.925	0.925	0.925	0.925	0.756	0.756	0.756	0.756	0.756

K	ρ_H	Prev.	p	Cen.	Comp.	Reg.	Ring	Hub	Cen.	Comp.	Reg.	Ring	Hub
10	0.0	0.30	2	0.940	0.940	0.940	0.940	0.940	0.909	0.909	0.909	0.909	0.909
10	0.0	0.07	2	0.927	0.927	0.927	0.927	0.927	0.876	0.876	0.876	0.876	0.876
10	0.1	0.30	2	0.945	0.945	0.945	0.945	0.945	0.931	0.931	0.931	0.931	0.931
10	0.1	0.07	2	0.920	0.920	0.920	0.920	0.920	0.846	0.846	0.846	0.846	0.846
10	0.2	0.30	2	0.900	0.900	0.900	0.900	0.900	0.789	0.789	0.789	0.789	0.789
10	0.2	0.07	2	0.930	0.930	0.930	0.930	0.930	0.880	0.880	0.879	0.880	0.880
15	0.0	0.30	2	0.935	0.935	0.935	0.935	0.935	0.952	0.952	0.952	0.952	0.951
15	0.0	0.07	2	0.953	0.953	0.953	0.953	0.953	1.006	1.006	1.006	1.006	1.005
15	0.1	0.30	2	0.935	0.935	0.935	0.935	0.935	0.894	0.894	0.894	0.894	0.894
15	0.1	0.07	2	0.902	0.902	0.902	0.902	0.902	0.886	0.886	0.885	0.886	0.885
15	0.2	0.30	2	0.965	0.965	0.965	0.965	0.965	0.998	0.998	0.998	0.998	0.998
15	0.2	0.07	2	0.935	0.935	0.935	0.935	0.935	0.839	0.839	0.839	0.839	0.838
15	0.0	0.30	3	0.950	0.950	0.950	0.950	0.950	0.968	0.968	0.968	0.968	0.968
15	0.1	0.30	3	0.945	0.945	0.945	0.945	0.945	0.918	0.918	0.918	0.918	0.917
15	0.2	0.30	3	0.920	0.920	0.920	0.920	0.920	0.908	0.908	0.908	0.908	0.907
25	0.0	0.30	2	0.943	0.943	0.943	0.938	0.940	0.930	0.930	0.930	0.915	0.913
25	0.0	0.07	2	0.935	0.935	0.935	0.935	0.932	0.897	0.897	0.896	0.884	0.881
25	0.1	0.30	2	0.955	0.955	0.955	0.953	0.950	1.006	1.006	1.005	0.992	0.988
25	0.1	0.07	2	0.945	0.945	0.945	0.940	0.943	0.965	0.965	0.964	0.952	0.946
25	0.2	0.30	2	0.970	0.970	0.970	0.970	0.960	0.995	0.995	0.995	0.980	0.978
25	0.2	0.07	2	0.925	0.925	0.925	0.920	0.920	0.942	0.942	0.942	0.924	0.924
25	0.0	0.30	3	0.960	0.960	0.960	0.958	0.958	1.002	1.002	1.001	0.993	0.982
25	0.1	0.30	3	0.955	0.955	0.955	0.950	0.953	0.992	0.992	0.992	0.981	0.976
25	0.2	0.30	3	0.915	0.915	0.915	0.905	0.910	0.845	0.845	0.845	0.831	0.828
50	0.0	0.30	2	0.958	0.958	0.958	0.945	0.912	1.012	1.012	1.011	0.984	0.838
50	0.0	0.07	2	0.953	0.953	0.953	0.948	0.895	1.011	1.011	1.010	0.982	0.836
50	0.1	0.30	2	0.927	0.927	0.930	0.930	0.885	0.990	0.990	0.989	0.958	0.829
50	0.1	0.07	2	0.935	0.935	0.935	0.920	0.877	0.922	0.922	0.921	0.889	0.758
50	0.2	0.30	2	0.920	0.920	0.920	0.920	0.865	0.926	0.926	0.924	0.883	0.772
50	0.2	0.07	2	0.945	0.945	0.945	0.925	0.840	0.946	0.946	0.945	0.897	0.790
50	0.0	0.30	3	0.948	0.948	0.948	0.943	0.885	0.996	0.996	0.995	0.970	0.825
50	0.1	0.30	3	0.927	0.927	0.927	0.912	0.850	0.943	0.943	0.942	0.907	0.790
50	0.2	0.30	3	0.935	0.935	0.935	0.870	0.795	0.944	0.944	0.943	0.894	0.800
100	0.0	0.30	2	0.935	0.935	0.935	0.925	0.562	0.970	0.970	0.962	0.928	0.514
100	0.0	0.07	2	0.960	0.960	0.958	0.953	0.555	1.015	1.015	1.008	0.974	0.521
100	0.1	0.30	2	0.958	0.958	0.955	0.940	0.362	1.011	1.011	1.004	0.970	0.537
100	0.1	0.07	2	0.950	0.950	0.948	0.922	0.482	0.997	0.997	0.990	0.950	0.553
100	0.2	0.30	2	0.955	0.955	0.955	0.940	0.200	0.988	0.988	0.981	0.940	0.563

K	ρ_H	Prev.	p	Cen.	Comp.	Reg.	Ring	Hub	Cen.	Comp.	Reg.	Ring	Hub
100	0.2	0.07	2	0.925	0.925	0.915	0.905	0.335	0.969	0.969	0.961	0.932	0.558
100	0.0	0.30	3	0.945	0.945	0.943	0.930	0.580	0.997	0.997	0.991	0.961	0.527
100	0.1	0.30	3	0.940	0.940	0.940	0.912	0.265	0.989	0.989	0.983	0.950	0.545
100	0.2	0.30	3	0.965	0.965	0.965	0.920	0.060	1.047	1.047	1.041	1.026	0.630

J Full simulation results

Tables S3 and S4 report every scenario of the simulation study in the main text, without averaging. Prev. is the average outcome prevalence (0.07, 0.30, and 0.50 for $\beta_0 = -3, -1, 0$). CV is the coefficient of variation of site size. t_{BM} is the t reference on score-space Bell–McCaffrey degrees of freedom. t_{FG} uses the degrees of freedom of Fay and Graubard. t_ν uses the effective-contributor count $(\sum_i q_{ij})^2 / \sum_i q_{ij}^2$, where q_{ij} is site i ’s contribution to the corrected variance of coefficient j . It is shown for comparison. Its coverage was at least 0.95 in every scenario, because it uses the realized scores and understates the degrees of freedom. ν^{BM} and ν^{FG} are median degrees of freedom. All 117,000 replicates completed.

Table S3: Coverage of nominal 95% intervals for β_x in all 117 scenarios. Each scenario had 1,000 replicates; Monte Carlo standard error about 0.007.

K	p	Prev.	CV	Uncorrected		KC			FG		MD	KC	ν^{BM}	ν^{FG}
				z	t_{BM}	t_{K-1}	t_{K-p}	t_{BM}	t_{BM}	t_{FG}	t_{BM}	t_ν		
10	2	0.50	0.30	0.883	0.927	0.937	0.941	0.942	0.942	0.943	0.950	0.965	8.1	8.0
10	2	0.50	0.78	0.876	0.932	0.933	0.934	0.944	0.944	0.950	0.963	0.966	5.9	5.4
10	2	0.50	1.00	0.876	0.935	0.930	0.934	0.958	0.957	0.964	0.976	0.966	5.0	4.5
10	2	0.30	0.30	0.909	0.943	0.950	0.951	0.952	0.952	0.953	0.963	0.975	8.1	8.0
10	2	0.30	0.78	0.877	0.939	0.938	0.941	0.956	0.956	0.961	0.970	0.972	5.9	5.4
10	2	0.30	1.00	0.875	0.938	0.931	0.936	0.955	0.955	0.961	0.972	0.964	5.0	4.5
10	2	0.07	0.30	0.882	0.934	0.942	0.944	0.943	0.944	0.946	0.960	0.975	7.9	7.6
10	2	0.07	0.78	0.874	0.940	0.935	0.948	0.959	0.960	0.966	0.976	0.981	5.7	5.3
10	2	0.07	1.00	0.867	0.949	0.941	0.945	0.965	0.965	0.970	0.977	0.973	4.9	4.4
10	5	0.50	0.30	0.903	0.938	0.941	0.964	0.947	0.944	0.944	0.956	0.970	8.1	8.0
10	5	0.50	0.78	0.892	0.938	0.938	0.968	0.959	0.959	0.965	0.975	0.978	5.9	5.4
10	5	0.50	1.00	0.881	0.945	0.943	0.966	0.969	0.970	0.974	0.983	0.969	5.0	4.4
10	5	0.30	0.30	0.909	0.939	0.950	0.973	0.954	0.955	0.956	0.966	0.981	8.1	7.9
10	5	0.30	0.78	0.876	0.933	0.934	0.957	0.951	0.947	0.951	0.962	0.969	5.8	5.4
10	5	0.30	1.00	0.876	0.947	0.947	0.965	0.967	0.967	0.971	0.978	0.971	5.0	4.4
10	5	0.07	0.30	0.893	0.936	0.939	0.969	0.947	0.949	0.948	0.962	0.974	7.7	7.5

K	p	Prev.	CV	z	t_{BM}	t_{K-1}	t_{K-p}	t_{BM}	t_{BM}	t_{FG}	t_{BM}	t_{ν}	ν^{BM}	ν^{FG}
10	5	0.07	0.78	0.882	0.941	0.936	0.963	0.962	0.964	0.964	0.972	0.973	5.7	5.1
10	5	0.07	1.00	0.863	0.929	0.924	0.954	0.954	0.959	0.967	0.974	0.971	4.9	4.3
18	2	0.50	0.30	0.922	0.943	0.947	0.947	0.947	0.948	0.948	0.955	0.973	15.3	15.2
18	2	0.50	0.78	0.892	0.930	0.932	0.933	0.943	0.942	0.945	0.951	0.959	10.9	10.3
18	2	0.50	1.00	0.925	0.948	0.948	0.948	0.963	0.960	0.967	0.974	0.974	9.0	8.3
18	2	0.30	0.30	0.917	0.937	0.945	0.946	0.949	0.949	0.949	0.959	0.975	15.3	15.1
18	2	0.30	0.78	0.900	0.939	0.939	0.941	0.951	0.951	0.953	0.959	0.968	10.9	10.3
18	2	0.30	1.00	0.918	0.942	0.939	0.939	0.952	0.952	0.954	0.964	0.967	9.0	8.3
18	2	0.07	0.30	0.924	0.941	0.945	0.946	0.949	0.950	0.950	0.958	0.971	14.7	14.5
18	2	0.07	0.78	0.911	0.942	0.938	0.940	0.951	0.952	0.957	0.960	0.969	10.6	10.0
18	2	0.07	1.00	0.909	0.943	0.940	0.941	0.951	0.951	0.955	0.962	0.964	8.9	8.2
18	5	0.50	0.30	0.942	0.953	0.955	0.956	0.956	0.956	0.956	0.960	0.976	15.3	15.2
18	5	0.50	0.78	0.920	0.946	0.946	0.954	0.957	0.958	0.959	0.966	0.971	10.9	10.3
18	5	0.50	1.00	0.903	0.931	0.931	0.935	0.945	0.945	0.949	0.954	0.961	9.0	8.4
18	5	0.30	0.30	0.922	0.945	0.955	0.961	0.957	0.957	0.957	0.965	0.976	15.3	15.1
18	5	0.30	0.78	0.912	0.942	0.946	0.951	0.955	0.955	0.956	0.968	0.971	10.9	10.2
18	5	0.30	1.00	0.891	0.929	0.928	0.930	0.935	0.939	0.944	0.955	0.965	8.9	8.3
18	5	0.07	0.30	0.912	0.930	0.939	0.947	0.943	0.944	0.944	0.952	0.971	14.7	14.4
18	5	0.07	0.78	0.905	0.937	0.938	0.943	0.945	0.948	0.950	0.956	0.971	10.6	9.9
18	5	0.07	1.00	0.897	0.922	0.920	0.924	0.939	0.941	0.944	0.958	0.959	8.8	7.9
18	10	0.50	0.30	0.928	0.956	0.959	0.971	0.959	0.959	0.959	0.963	0.977	15.3	15.2
18	10	0.50	0.78	0.920	0.941	0.942	0.952	0.948	0.948	0.949	0.955	0.962	10.7	10.2
18	10	0.50	1.00	0.906	0.938	0.934	0.952	0.953	0.952	0.955	0.960	0.966	9.0	8.4
18	10	0.30	0.30	0.933	0.952	0.956	0.970	0.958	0.958	0.958	0.965	0.975	15.3	15.1
18	10	0.30	0.78	0.921	0.950	0.952	0.966	0.957	0.956	0.960	0.967	0.975	10.9	10.4
18	10	0.30	1.00	0.902	0.933	0.935	0.952	0.949	0.946	0.949	0.963	0.964	9.0	8.4
18	10	0.07	0.30	0.920	0.954	0.958	0.970	0.959	0.959	0.959	0.969	0.974	14.6	14.3
18	10	0.07	0.78	0.895	0.928	0.929	0.948	0.943	0.944	0.944	0.955	0.961	10.4	9.8
18	10	0.07	1.00	0.876	0.920	0.914	0.935	0.935	0.938	0.946	0.955	0.954	8.8	8.0
30	2	0.50	0.30	0.936	0.947	0.950	0.951	0.952	0.952	0.952	0.956	0.970	26.2	26.0
30	2	0.50	0.78	0.935	0.951	0.952	0.952	0.956	0.956	0.956	0.960	0.968	18.2	17.6
30	2	0.50	1.00	0.926	0.944	0.944	0.945	0.951	0.951	0.951	0.958	0.963	14.9	14.2
30	2	0.30	0.30	0.930	0.944	0.946	0.946	0.946	0.947	0.947	0.953	0.970	26.1	25.9
30	2	0.30	0.78	0.924	0.937	0.937	0.937	0.943	0.943	0.943	0.949	0.960	18.3	17.6
30	2	0.30	1.00	0.918	0.939	0.936	0.936	0.947	0.947	0.947	0.953	0.962	15.0	14.2
30	2	0.07	0.30	0.938	0.946	0.950	0.951	0.953	0.953	0.953	0.957	0.968	25.2	24.9
30	2	0.07	0.78	0.932	0.940	0.942	0.943	0.949	0.949	0.950	0.961	0.967	17.7	17.1
30	2	0.07	1.00	0.920	0.942	0.939	0.940	0.949	0.949	0.950	0.956	0.964	14.7	13.9
30	5	0.50	0.30	0.932	0.947	0.948	0.950	0.949	0.950	0.950	0.951	0.960	26.2	26.0
30	5	0.50	0.78	0.925	0.940	0.942	0.944	0.948	0.948	0.948	0.950	0.963	18.3	17.6

K	p	Prev.	CV	z	t_{BM}	t_{K-1}	t_{K-p}	t_{BM}	t_{BM}	t_{FG}	t_{BM}	t_{ν}	ν^{BM}	ν^{FG}
30	5	0.50	1.00	0.910	0.936	0.936	0.937	0.947	0.947	0.947	0.952	0.954	15.2	14.4
30	5	0.30	0.30	0.928	0.935	0.940	0.941	0.940	0.940	0.940	0.945	0.953	26.0	25.9
30	5	0.30	0.78	0.910	0.930	0.929	0.932	0.939	0.939	0.941	0.946	0.962	18.2	17.5
30	5	0.30	1.00	0.917	0.938	0.937	0.939	0.946	0.946	0.947	0.955	0.963	14.9	14.1
30	5	0.07	0.30	0.913	0.931	0.936	0.938	0.938	0.939	0.939	0.944	0.963	25.0	24.7
30	5	0.07	0.78	0.910	0.925	0.924	0.926	0.932	0.931	0.935	0.945	0.965	17.8	17.1
30	5	0.07	1.00	0.927	0.945	0.943	0.944	0.949	0.948	0.948	0.954	0.965	14.6	13.7
30	10	0.50	0.30	0.933	0.942	0.943	0.947	0.944	0.943	0.943	0.949	0.960	26.1	26.0
30	10	0.50	0.78	0.924	0.944	0.944	0.951	0.952	0.952	0.952	0.960	0.971	18.5	17.9
30	10	0.50	1.00	0.927	0.950	0.949	0.951	0.957	0.957	0.959	0.967	0.976	15.1	14.3
30	10	0.30	0.30	0.930	0.939	0.942	0.948	0.942	0.942	0.942	0.950	0.964	26.1	25.9
30	10	0.30	0.78	0.925	0.937	0.942	0.948	0.951	0.949	0.951	0.959	0.970	18.3	17.6
30	10	0.30	1.00	0.924	0.940	0.941	0.942	0.947	0.949	0.951	0.959	0.963	15.2	14.4
30	10	0.07	0.30	0.924	0.934	0.941	0.943	0.943	0.943	0.943	0.946	0.959	25.1	24.8
30	10	0.07	0.78	0.926	0.937	0.937	0.941	0.942	0.943	0.945	0.950	0.958	17.6	16.9
30	10	0.07	1.00	0.929	0.949	0.947	0.953	0.957	0.955	0.956	0.963	0.965	14.7	13.8
30	20	0.50	0.30	0.913	0.935	0.938	0.958	0.939	0.938	0.938	0.943	0.954	26.1	26.0
30	20	0.50	0.78	0.927	0.941	0.942	0.964	0.949	0.948	0.950	0.960	0.965	18.2	17.6
30	20	0.50	1.00	0.913	0.934	0.933	0.948	0.938	0.939	0.939	0.949	0.953	15.0	14.2
30	20	0.30	0.30	0.919	0.938	0.942	0.965	0.944	0.944	0.944	0.949	0.966	26.1	25.9
30	20	0.30	0.78	0.935	0.956	0.954	0.973	0.964	0.961	0.962	0.970	0.972	18.2	17.6
30	20	0.30	1.00	0.927	0.953	0.952	0.964	0.960	0.960	0.960	0.964	0.971	15.0	14.2
30	20	0.07	0.30	0.918	0.927	0.929	0.952	0.932	0.930	0.930	0.938	0.951	25.1	24.7
30	20	0.07	0.78	0.925	0.940	0.942	0.956	0.947	0.946	0.946	0.954	0.963	17.6	16.9
30	20	0.07	1.00	0.913	0.930	0.931	0.947	0.941	0.944	0.948	0.954	0.952	14.6	13.7
50	2	0.50	0.30	0.937	0.944	0.946	0.946	0.946	0.946	0.946	0.950	0.957	44.2	44.0
50	2	0.50	0.78	0.952	0.957	0.958	0.958	0.963	0.963	0.963	0.966	0.969	30.7	30.0
50	2	0.50	1.00	0.946	0.955	0.954	0.955	0.962	0.962	0.963	0.966	0.969	24.9	24.1
50	2	0.30	0.30	0.941	0.949	0.951	0.951	0.952	0.952	0.952	0.953	0.959	44.1	43.9
50	2	0.30	0.78	0.942	0.953	0.952	0.952	0.955	0.955	0.955	0.958	0.963	30.5	29.8
50	2	0.30	1.00	0.932	0.945	0.944	0.944	0.952	0.952	0.952	0.957	0.969	24.9	24.0
50	2	0.07	0.30	0.934	0.939	0.941	0.941	0.941	0.941	0.941	0.942	0.954	42.5	42.3
50	2	0.07	0.78	0.912	0.924	0.925	0.926	0.929	0.929	0.930	0.941	0.956	29.7	28.9
50	2	0.07	1.00	0.939	0.951	0.950	0.950	0.955	0.956	0.957	0.961	0.967	24.5	23.6
50	5	0.50	0.30	0.940	0.946	0.950	0.950	0.950	0.950	0.950	0.952	0.960	44.2	44.0
50	5	0.50	0.78	0.947	0.953	0.954	0.955	0.959	0.958	0.959	0.960	0.969	30.6	30.0
50	5	0.50	1.00	0.933	0.943	0.944	0.944	0.951	0.950	0.952	0.958	0.971	25.0	24.1
50	5	0.30	0.30	0.947	0.953	0.954	0.955	0.955	0.955	0.955	0.956	0.961	44.0	43.8
50	5	0.30	0.78	0.945	0.962	0.965	0.965	0.971	0.970	0.970	0.975	0.980	30.6	30.0
50	5	0.30	1.00	0.918	0.930	0.929	0.931	0.941	0.941	0.941	0.947	0.958	25.0	24.2

K	p	Prev.	CV	z	t_{BM}	t_{K-1}	t_{K-p}	t_{BM}	t_{BM}	t_{FG}	t_{BM}	t_{ν}	ν^{BM}	ν^{FG}
50	5	0.07	0.30	0.945	0.948	0.952	0.953	0.953	0.953	0.953	0.955	0.967	42.4	42.1
50	5	0.07	0.78	0.928	0.936	0.936	0.937	0.943	0.942	0.942	0.946	0.960	29.7	28.9
50	5	0.07	1.00	0.929	0.940	0.940	0.940	0.945	0.944	0.944	0.951	0.960	24.2	23.3
50	10	0.50	0.30	0.941	0.949	0.949	0.949	0.949	0.949	0.949	0.952	0.959	44.2	44.0
50	10	0.50	0.78	0.940	0.950	0.951	0.952	0.953	0.953	0.953	0.956	0.959	30.7	30.0
50	10	0.50	1.00	0.928	0.940	0.940	0.940	0.944	0.944	0.944	0.951	0.961	25.2	24.3
50	10	0.30	0.30	0.944	0.949	0.950	0.950	0.950	0.950	0.950	0.953	0.963	44.0	43.8
50	10	0.30	0.78	0.932	0.937	0.938	0.938	0.940	0.940	0.940	0.946	0.954	30.6	29.9
50	10	0.30	1.00	0.938	0.947	0.947	0.947	0.949	0.950	0.950	0.953	0.965	25.0	24.1
50	10	0.07	0.30	0.942	0.950	0.953	0.954	0.954	0.954	0.954	0.956	0.968	42.4	42.2
50	10	0.07	0.78	0.944	0.952	0.952	0.953	0.955	0.955	0.956	0.961	0.973	29.8	29.0
50	10	0.07	1.00	0.928	0.937	0.937	0.937	0.941	0.943	0.944	0.948	0.960	24.5	23.4
50	20	0.50	0.30	0.942	0.950	0.953	0.955	0.953	0.953	0.953	0.955	0.960	44.2	44.0
50	20	0.50	0.78	0.944	0.952	0.952	0.953	0.954	0.954	0.954	0.954	0.962	30.4	29.7
50	20	0.50	1.00	0.914	0.927	0.928	0.937	0.939	0.938	0.939	0.946	0.957	25.2	24.3
50	20	0.30	0.30	0.940	0.944	0.946	0.950	0.946	0.946	0.946	0.950	0.953	44.0	43.8
50	20	0.30	0.78	0.932	0.943	0.944	0.946	0.949	0.947	0.948	0.949	0.959	30.5	29.8
50	20	0.30	1.00	0.934	0.945	0.944	0.946	0.949	0.949	0.949	0.952	0.956	25.1	24.2
50	20	0.07	0.30	0.945	0.951	0.953	0.958	0.954	0.954	0.954	0.958	0.963	42.3	41.9
50	20	0.07	0.78	0.927	0.938	0.938	0.944	0.944	0.944	0.944	0.949	0.958	29.6	28.9
50	20	0.07	1.00	0.924	0.941	0.940	0.942	0.943	0.942	0.942	0.952	0.959	24.2	23.3

Table S4: Standard-error calibration (SE/SD) by variance estimator, and Fed-GEE (KC with t_{BM}) compared with meta-analytic GEE, in all 117 scenarios.

K	p	Prev.	CV	SE/SD				Fed-GEE		Meta-analytic GEE	
				Unc.	KC	MD	FG	Cov.	Bias/SD	Cov.	Bias/SD
10	2	0.50	0.30	0.894	0.949	1.008	0.947	0.942	0.10	0.912	0.23
10	2	0.50	0.78	0.860	0.932	1.016	0.931	0.944	0.12	0.893	0.17
10	2	0.50	1.00	0.862	0.946	1.049	0.945	0.958	0.08	0.890	0.18
10	2	0.30	0.30	0.918	0.975	1.036	0.974	0.952	0.11	0.909	0.19
10	2	0.30	0.78	0.854	0.927	1.012	0.926	0.956	0.07	0.875	0.07
10	2	0.30	1.00	0.813	0.894	0.991	0.893	0.955	0.07	0.854	0.07
10	2	0.07	0.30	0.865	0.920	0.980	0.923	0.943	0.12	0.726	-0.31
10	2	0.07	0.78	0.846	0.920	1.005	0.924	0.959	0.13	0.723	-0.25
10	2	0.07	1.00	0.822	0.907	1.012	0.912	0.965	0.12	0.733	-0.22
10	5	0.50	0.30	0.924	0.983	1.048	0.980	0.947	0.22	0.876	0.51
10	5	0.50	0.78	0.882	0.958	1.047	0.955	0.959	0.18	0.884	0.39

K	p	Prev.	CV	Unc.	KC	MD	FG	Cov.	Bias/SD	Cov.	Bias/SD
10	5	0.50	1.00	0.880	0.969	1.078	0.967	0.969	0.16	0.882	0.39
10	5	0.30	0.30	0.928	0.988	1.054	0.987	0.954	0.16	0.892	0.36
10	5	0.30	0.78	0.852	0.928	1.016	0.928	0.951	0.16	0.870	0.33
10	5	0.30	1.00	0.866	0.956	1.068	0.955	0.967	0.21	0.867	0.36
10	5	0.07	0.30	0.890	0.951	1.020	0.952	0.947	0.13	0.787	-0.12
10	5	0.07	0.78	0.855	0.934	1.028	0.939	0.962	0.12	0.818	-0.06
10	5	0.07	1.00	0.824	0.913	1.024	0.920	0.954	0.17	0.818	0.02
18	2	0.50	0.30	0.940	0.972	1.004	0.970	0.947	0.10	0.888	0.27
18	2	0.50	0.78	0.870	0.910	0.953	0.909	0.943	0.08	0.885	0.20
18	2	0.50	1.00	0.953	1.006	1.066	1.005	0.963	0.09	0.865	0.19
18	2	0.30	0.30	0.927	0.958	0.990	0.957	0.949	0.05	0.900	0.16
18	2	0.30	0.78	0.908	0.950	0.996	0.949	0.951	0.08	0.873	0.06
18	2	0.30	1.00	0.941	0.994	1.053	0.994	0.952	0.10	0.859	0.05
18	2	0.07	0.30	0.938	0.970	1.004	0.972	0.949	0.07	0.641	-0.54
18	2	0.07	0.78	0.906	0.949	0.996	0.951	0.951	0.03	0.654	-0.52
18	2	0.07	1.00	0.893	0.944	1.000	0.945	0.951	0.09	0.686	-0.44
18	5	0.50	0.30	1.002	1.037	1.073	1.034	0.956	0.14	0.876	0.60
18	5	0.50	0.78	0.947	0.993	1.043	0.990	0.957	0.14	0.867	0.50
18	5	0.50	1.00	0.874	0.924	0.980	0.923	0.945	0.15	0.836	0.46
18	5	0.30	0.30	1.007	1.042	1.078	1.040	0.957	0.11	0.887	0.48
18	5	0.30	0.78	0.946	0.992	1.041	0.991	0.955	0.18	0.856	0.46
18	5	0.30	1.00	0.894	0.946	1.005	0.947	0.935	0.18	0.849	0.41
18	5	0.07	0.30	0.918	0.952	0.988	0.951	0.943	0.12	0.748	-0.22
18	5	0.07	0.78	0.923	0.970	1.022	0.972	0.945	0.13	0.803	-0.15
18	5	0.07	1.00	0.871	0.922	0.981	0.927	0.939	0.11	0.817	-0.13
18	10	0.50	0.30	0.978	1.014	1.051	1.009	0.959	0.19	0.618	1.37
18	10	0.50	0.78	0.916	0.961	1.011	0.957	0.948	0.18	0.699	0.99
18	10	0.50	1.00	0.918	0.972	1.033	0.969	0.953	0.22	0.735	1.00
18	10	0.30	0.30	0.976	1.012	1.049	1.009	0.958	0.19	0.685	1.22
18	10	0.30	0.78	0.982	1.031	1.085	1.027	0.957	0.23	0.757	1.03
18	10	0.30	1.00	0.888	0.939	0.998	0.937	0.949	0.19	0.757	0.86
18	10	0.07	0.30	0.941	0.979	1.020	0.976	0.959	0.16	0.800	0.29
18	10	0.07	0.78	0.884	0.933	0.986	0.931	0.943	0.13	0.845	0.31
18	10	0.07	1.00	0.852	0.905	0.966	0.910	0.935	0.21	0.844	0.29
30	2	0.50	0.30	0.996	1.016	1.036	1.015	0.952	-0.01	0.912	0.28
30	2	0.50	0.78	0.973	1.000	1.029	1.000	0.956	0.08	0.883	0.25
30	2	0.50	1.00	0.940	0.972	1.007	0.971	0.951	0.07	0.855	0.19
30	2	0.30	0.30	0.956	0.975	0.994	0.975	0.946	0.06	0.895	0.21
30	2	0.30	0.78	0.917	0.943	0.970	0.942	0.943	0.06	0.862	0.07
30	2	0.30	1.00	0.935	0.968	1.002	0.967	0.947	0.05	0.851	0.04

K	p	Prev.	CV	Unc.	KC	MD	FG	Cov.	Bias/SD	Cov.	Bias/SD
30	2	0.07	0.30	0.970	0.989	1.009	0.990	0.953	0.04	0.543	-0.76
30	2	0.07	0.78	0.925	0.951	0.979	0.953	0.949	0.06	0.592	-0.63
30	2	0.07	1.00	0.920	0.951	0.985	0.953	0.949	0.09	0.633	-0.59
30	5	0.50	0.30	0.959	0.978	0.999	0.977	0.949	0.07	0.841	0.69
30	5	0.50	0.78	0.947	0.974	1.003	0.973	0.948	0.06	0.836	0.55
30	5	0.50	1.00	0.884	0.915	0.948	0.913	0.947	0.15	0.779	0.57
30	5	0.30	0.30	0.943	0.963	0.983	0.961	0.940	0.12	0.842	0.58
30	5	0.30	0.78	0.951	0.978	1.007	0.977	0.939	0.14	0.844	0.52
30	5	0.30	1.00	0.939	0.972	1.007	0.970	0.946	0.10	0.831	0.43
30	5	0.07	0.30	0.961	0.982	1.004	0.981	0.938	0.04	0.742	-0.38
30	5	0.07	0.78	0.910	0.937	0.966	0.938	0.932	0.08	0.770	-0.24
30	5	0.07	1.00	0.924	0.958	0.995	0.959	0.949	0.04	0.760	-0.27
30	10	0.50	0.30	0.971	0.992	1.013	0.990	0.944	0.11	0.489	1.64
30	10	0.50	0.78	0.952	0.980	1.009	0.977	0.952	0.18	0.580	1.42
30	10	0.50	1.00	0.982	1.018	1.057	1.015	0.957	0.16	0.634	1.31
30	10	0.30	0.30	0.972	0.992	1.014	0.990	0.942	0.14	0.531	1.58
30	10	0.30	0.78	0.941	0.970	1.000	0.967	0.951	0.15	0.621	1.21
30	10	0.30	1.00	0.973	1.007	1.044	1.006	0.947	0.20	0.655	1.14
30	10	0.07	0.30	0.940	0.963	0.986	0.961	0.943	0.15	0.792	0.31
30	10	0.07	0.78	0.926	0.956	0.988	0.954	0.942	0.08	0.820	0.28
30	10	0.07	1.00	0.954	0.990	1.028	0.990	0.957	0.15	0.853	0.30
30	20	0.50	0.30	0.924	0.945	0.967	0.941	0.939	0.23	0.066	2.83
30	20	0.50	0.78	0.935	0.964	0.996	0.960	0.949	0.20	0.227	2.18
30	20	0.50	1.00	0.900	0.933	0.969	0.929	0.938	0.29	0.283	2.04
30	20	0.30	0.30	0.955	0.978	1.001	0.974	0.944	0.13	0.135	2.49
30	20	0.30	0.78	0.962	0.992	1.025	0.988	0.964	0.20	0.297	2.02
30	20	0.30	1.00	0.941	0.977	1.015	0.974	0.960	0.23	0.385	1.81
30	20	0.07	0.30	0.923	0.948	0.975	0.943	0.932	0.19	0.805	0.59
30	20	0.07	0.78	0.937	0.971	1.007	0.965	0.947	0.18	0.776	0.72
30	20	0.07	1.00	0.909	0.947	0.990	0.943	0.941	0.24	0.786	0.72
50	2	0.50	0.30	0.968	0.979	0.990	0.979	0.946	0.01	0.893	0.31
50	2	0.50	0.78	0.990	1.006	1.023	1.006	0.963	0.02	0.891	0.25
50	2	0.50	1.00	0.977	0.997	1.018	0.996	0.962	0.08	0.877	0.24
50	2	0.30	0.30	0.969	0.981	0.992	0.981	0.952	0.02	0.911	0.24
50	2	0.30	0.78	0.981	0.998	1.015	0.997	0.955	0.06	0.870	0.12
50	2	0.30	1.00	0.954	0.973	0.994	0.973	0.952	0.05	0.844	0.03
50	2	0.07	0.30	0.951	0.962	0.974	0.963	0.941	0.02	0.433	-1.07
50	2	0.07	0.78	0.926	0.942	0.958	0.943	0.929	0.06	0.507	-0.89
50	2	0.07	1.00	0.981	1.001	1.023	1.002	0.955	0.08	0.538	-0.85
50	5	0.50	0.30	1.014	1.027	1.039	1.026	0.950	-0.01	0.822	0.82

K	p	Prev.	CV	Unc.	KC	MD	FG	Cov.	Bias/SD	Cov.	Bias/SD
50	5	0.50	0.78	0.996	1.013	1.031	1.012	0.959	0.06	0.775	0.77
50	5	0.50	1.00	0.953	0.973	0.994	0.972	0.951	0.09	0.763	0.73
50	5	0.30	0.30	0.999	1.011	1.024	1.010	0.955	0.10	0.813	0.77
50	5	0.30	0.78	1.016	1.034	1.052	1.033	0.971	0.11	0.795	0.70
50	5	0.30	1.00	0.922	0.942	0.962	0.941	0.941	0.13	0.775	0.52
50	5	0.07	0.30	1.016	1.029	1.043	1.029	0.953	0.06	0.692	-0.48
50	5	0.07	0.78	0.928	0.945	0.962	0.945	0.943	0.06	0.733	-0.37
50	5	0.07	1.00	0.941	0.962	0.984	0.962	0.945	0.06	0.753	-0.35
50	10	0.50	0.30	0.976	0.988	1.001	0.987	0.949	0.12	0.301	2.11
50	10	0.50	0.78	0.967	0.984	1.002	0.982	0.953	0.17	0.414	1.75
50	10	0.50	1.00	0.953	0.974	0.995	0.972	0.944	0.20	0.473	1.65
50	10	0.30	0.30	0.974	0.987	1.000	0.986	0.950	0.07	0.396	1.88
50	10	0.30	0.78	0.940	0.957	0.975	0.955	0.940	0.15	0.482	1.57
50	10	0.30	1.00	0.958	0.979	1.001	0.978	0.949	0.14	0.559	1.42
50	10	0.07	0.30	0.991	1.005	1.019	1.004	0.954	0.10	0.787	0.36
50	10	0.07	0.78	0.980	0.999	1.018	0.998	0.955	0.08	0.822	0.35
50	10	0.07	1.00	0.941	0.962	0.985	0.962	0.941	0.11	0.817	0.40
50	20	0.50	0.30	0.985	0.998	1.012	0.996	0.953	0.16	0.010	3.68
50	20	0.50	0.78	0.976	0.994	1.014	0.992	0.954	0.15	0.073	2.84
50	20	0.50	1.00	0.927	0.947	0.969	0.945	0.939	0.17	0.132	2.50
50	20	0.30	0.30	0.970	0.984	0.997	0.981	0.946	0.18	0.028	3.15
50	20	0.30	0.78	0.977	0.996	1.015	0.993	0.949	0.19	0.110	2.72
50	20	0.30	1.00	0.926	0.947	0.969	0.945	0.949	0.15	0.190	2.32
50	20	0.07	0.30	0.982	0.998	1.015	0.995	0.954	0.13	0.740	0.70
50	20	0.07	0.78	0.945	0.965	0.986	0.962	0.944	0.13	0.727	0.82
50	20	0.07	1.00	0.930	0.953	0.978	0.951	0.943	0.22	0.711	0.92

K VHA application: additional results

Web Table S5 gives the VHA confidence intervals under the uncorrected sandwich with normal and Bell–McCaffrey references, the KC and FG corrections, FG with its own degrees of freedom, and CR2 with Satterthwaite degrees of freedom. The VISNs are nearly equal in size (CV 0.35), so the corrections change the standard errors by at most 12%. Every column gives the same conclusion for every coefficient.

Table S5: VHA application: 95% confidence intervals for the adjusted odds ratios under alternative variance corrections and reference distributions. All columns share the point estimates in Table 3 of the main text.

Covariate	Uncorrected		KC	FG		CR2
	z	t_{BM}	t_{BM}	t_{BM}	t_{FG}	Satterthwaite
<i>Demographics</i>						
Female	1.05, 1.41	1.04, 1.43	1.03, 1.44	1.03, 1.43	1.03, 1.43	1.03, 1.44
Age ≥ 65 years	0.79, 0.98	0.79, 0.99	0.78, 0.99	0.78, 0.99	0.78, 1.00	0.78, 0.99
<i>Alcohol severity</i>						
AUDIT-C no or low use	0.84, 0.93	0.84, 0.94	0.83, 0.94	0.84, 0.94	0.83, 0.94	0.83, 0.94
Opioid use disorder	0.54, 0.68	0.53, 0.69	0.53, 0.69	0.53, 0.69	0.53, 0.69	0.53, 0.69
<i>Medical complexity</i>						
CAN score ≥ 60 th percentile	0.65, 0.78	0.65, 0.79	0.64, 0.79	0.64, 0.80	0.64, 0.80	0.64, 0.79
Frailty, mild or worse	0.61, 0.78	0.60, 0.78	0.60, 0.79	0.60, 0.79	0.60, 0.79	0.60, 0.79
Any ICU stay	0.76, 1.02	0.75, 1.04	0.74, 1.05	0.74, 1.04	0.74, 1.04	0.74, 1.05
<i>Hospital process</i>						
Psychiatry discharging service	1.43, 2.09	1.41, 2.13	1.39, 2.15	1.39, 2.15	1.39, 2.15	1.39, 2.15
Addiction consult						
Clinician	1.19, 1.82	1.15, 1.87	1.14, 1.90	1.14, 1.90	1.14, 1.90	1.14, 1.90
Non-clinician	1.36, 2.59	1.29, 2.73	1.24, 2.84	1.24, 2.85	1.23, 2.87	1.24, 2.84
Discharge against advice	0.66, 1.92	0.58, 2.19	0.53, 2.41	0.53, 2.41	0.52, 2.46	0.53, 2.40

Uncorrected, VISN-clustered sandwich. KC, FG, score-space corrections. t_{BM} , t reference on score-space Bell–McCaffrey degrees of freedom; t_{FG} , degrees of freedom of Fay and Graubard. CR2, bias-reduced linearization with Satterthwaite degrees of freedom computed on the pooled data with the clubSandwich package. The KC, t_{BM} column is the recommended procedure. CAN, Care Assessment Need; ICU, intensive care unit; AUDIT-C, Alcohol Use Disorders Identification Test–Consumption. Reference categories: male; age 18–64 years; AUDIT-C unhealthy use or higher; no opioid use disorder; CAN score below the 60th percentile; nonfrail or prefrail; no ICU stay; medical or surgical discharging service; no addiction consult; routine discharge.

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